

Anaesthesia for minimally invasive cardiac surgery

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Minimally invasive cardiac surgery is used for both extracardiac and intracardiac procedures. Extracardiac procedures, such as coronary artery bypass grafting, are often performed on a beating heart. Intracardiac procedures are done with the aid of cardiopulmonary bypass. The surgery is performed via a minithoracotomy or a ministernotomy. Thoracoscopic video-assisted surgery, often with robotic assistance, necessitates prolonged one-lung ventilation to optimize exposure. Port-access surgery will require appropriate positioning of various catheters to establish cardiopulmonary bypass. Adequate flow during cardiopulmonary bypass may require suction augmentation of venous return and may increase the risk of air emboli. Limited exposure of the heart during surgery poses challenges with management of arrhythmia, haemostasis, myocardial protection and de-airing at the end of surgery. Patient selection is important to avoid intra-operative and post-operative complications. Prolonged single-lung ventilation, incomplete revascularization in hybrid procedures, and limited access for rapid intervention pose challenges with patient management. Conversion to sternotomy that may be required occasionally and extension of portals over several dermatomal segments mandate a versatile analgesic technique.

Key words: anaesthesia; regional; minimally invasive surgery; pain; stress response.

Currently, a number of cardiac procedures are performed as minimally invasive procedures. The factors driving the development of such techniques are improved cosmesis and healing, reduction in stress response, reduced stay in hospital and in the intensive care unit (ITU), and reduced transfusion requirements, all of which result in overall reduction in cost. Zenati et al¹ document that, compared to traditional coronary artery bypass grafting (CABG) performed with the aid of cardiopulmonary bypass (CPB), minimally invasive direct coronary artery bypass (MIDCAB) reduced stay in the intensive care unit (12.3 ± 3.3 hours versus 32.3 ± 12.6 hours), transfusion requirements (0 versus 42%), cost of total care (\$12 885 $\pm$ 1511 versus \$21 260 $\pm$ 5497) and length of stay in hospital (2.5 ± 0.8 versus 5.9 ± 2 days). A number of investigators have noted similar benefits with both CABG and minimally invasive valve surgery.^{2–5} Although aortic cross-clamp time is prolonged in minimally invasive surgery, Frazier et al⁶ note that patients returned to work earlier following partial sternotomy. While avoiding

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primary full sternotomy reduces the chances of substernal adhesions, avoiding redo sternotomy is particularly attractive in reducing blood loss and avoiding damage to the heart and pre-existing grafts. Repair of atrial septal defect, ligation of patent ductus arteriosus and, recently, ablative procedures for arrhythmias, are also performed using the minimal access technique with improved outcome. By and large, MIDCAB without CPB is done for single-vessel coronary artery disease.⁷ Occasionally it is performed as a hybrid procedure for multivessel disease where the culprit lesion is bypassed as MIDCAB and the other lesions are treated with endovascular techniques such as angioplasty and/or stents in the post-operative period.⁸

Cardiopulmonary bypass results in a cascade of cytokine response and stress hormone release contributing to the multisystem organ failure in the post-operative period.^{9–11} By avoiding cardiopulmonary bypass, MIDCAB results in reduced surgical stress,¹² less fluid shift, fewer neuropsychological disturbances,¹³ and avoidance of hypothermia and associated complications. Anaesthesia and analgesia should be tailored to accomplish the benefits of such innovations in surgery.

SURGICAL APPROACH

MIDCAB

Although the original description of minimally invasive coronary bypass involved direct dissection of internal mammary artery that was anastomosed to the target coronary artery, it has undergone several stages of refinement in the last several years (Table I). The surgery may be performed using a variety of surgical approaches such as left or right anterior minithoracotomy, upper or lower ministernotomy or an epigastric approach.⁷ Endoscopic harvesting of internal mammary artery (IMA) as well as saphenous vein is done prior to the main surgical exposure for anastomosis.¹⁴ Thoracoscopic harvesting of IMA requires ports for camera, harmonic scalpel and insufflator that are inserted via different intercostal spaces, usually T3 to T6 (Figure I). Occasionally radial artery or

Table I. Minimally invasive cardiac surgical techniques.

<i>Procedures performed on the beating heart</i>
CABG
MIDCAB
Video-assisted direct CAB
Totally endoscopic CAB (TECAB)
PADCAB (perfusion-assisted direct CAB)
Robotic-assisted video-enhanced CAB (RAVECAB)
Other procedures
Procedures to treat arrhythmias
Closure of PDA
<i>Procedures performed on the arrested heart</i>
Port-access CAB
RAVECAB
Port-access valve replacements
Port-access valve repairs
Direct-access valve procedures
Closure of ASD
Excision of atrial myxoma
ASD; atrial septal defect.

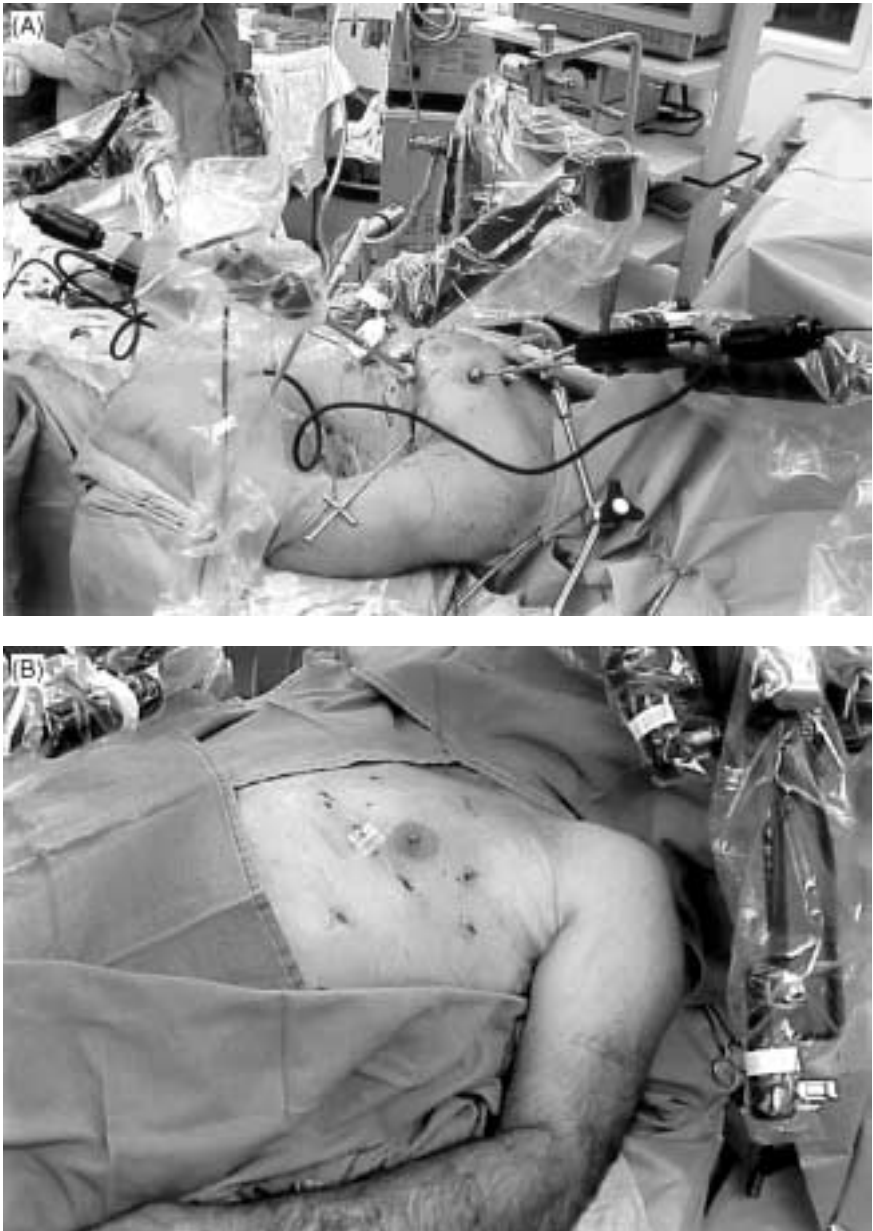


Figure 1. (A) Multiple ports via several intercostal spaces are used to facilitate thoracoscopic surgery. During dissection of the internal mammary artery, the harmonic scalpel and scope are inserted in the midaxillary line as high as the 3rd intercostal space. (B) Shows the multiple portals of entry extending over several intercostal spaces at the end of surgery.

saphenous vein grafts may be required to extend the length of IMA or to form a bifurcation 'H' or 'Y' conduit. Gastro-epiploic artery is being used to revascularize the right coronary or the posterior circulation.¹⁵ These patients have an upper abdominal

incision with extension of the area of surgical pain. Anastomosis involves the use of a stabilizer, retractor and proximal and distal silastic sutures to control back bleeding. Intracoronary shunts are often used to facilitate anastomosis and to maintain some distal coronary perfusion. A variety of stabilizers are used^{16,17} that provide a relatively motionless field to perform anastomosis on a beating heart, but a steady heart rate without irritability improves surgical conditions. The ipsilateral lung is collapsed during both dissection of the IMA and coronary anastomosis to avoid movement induced by respiratory excursions. Currently, the surgery often involves use of technology in development and is usually prolonged. In our experience the average duration of surgery is 262 ± 60 minutes when the anastomosis is performed via minithoracotomy and significantly longer if anastomosis is performed using robotic assistance. Robotic assisted anastomosis with virtual cardiac image arrest is just undergoing evaluation. Conversion to sternotomy and/or cardiopulmonary bypass is required in 7–20% of patients. Factors that contribute to conversion include poor visualization of IMA due to fat, abnormal coronary anatomy such as an intramyocardial course of the left anterior descending artery, calcification of the coronary artery, abnormal chest contour and accidental trauma to the ventricle or to the coronary artery by the stabilizer.

Minimally invasive intracardiac surgery

Both mitral and aortic valve surgeries, ascending aortic root surgery, closure of atrial septal defect and removal of atrial myxoma, using the minimally invasive approach, are done either via the right lateral minithoracotomy or upper ministernotomy.¹⁸ Cardiopulmonary bypass is initiated via the femoral vessels assisted by suction (–20 to –40 mmHg) to augment venous return. Once the thoracic cavity is exposed, direct cannulation of the aorta may be done (i) to deliver antegrade cardioplegia and (ii) for arterial return from the pump. The vena cava may be cannulated to augment venous return to the CPB circuit. A coronary sinus catheter is inserted using echocardiographic guidance, and retrograde cardioplegia is used where indicated. De-airing at the end of surgery is done via the aortic root with echocardiographic assistance. Percutaneous cardiopulmonary bypass may also be initiated using the port-access devices¹⁹ (Figure 2). The catheters are positioned percutaneously by the anaesthesiologist using echocardiography and/or fluoroscopy.^{20,21} Ongoing monitoring during surgery is essential to ensure that the endo-aortic balloon clamp has not dislodged to occlude the carotids.²² Although the initial experience documented improved outcome with patients,²³ Chaney et al²⁴ in their series note that port-access surgery did not result in earlier extubation, or shorter ICU stay, and necessitated prolongation of cross-clamp and pump time. This approach did result in a shorter hospital stay in patients who had CABG. Other potentially catastrophic complications that the anaesthesiologist should be aware of with this device include cardiac perforation and rupture of the coronary sinus.^{21,25}

PATIENT SELECTION AND PREPARATION

The surgical technique, limited access to surgical site, anaesthetic requirement for prolonged one-lung ventilation, surgical incision extending over several dermatomal segments, post-operative pain management and the potential for incomplete revascularization in some patients undergoing MIDCAB make patient selection crucial. Ideally these patients should be seen in the pre-operative anaesthesia clinic by an

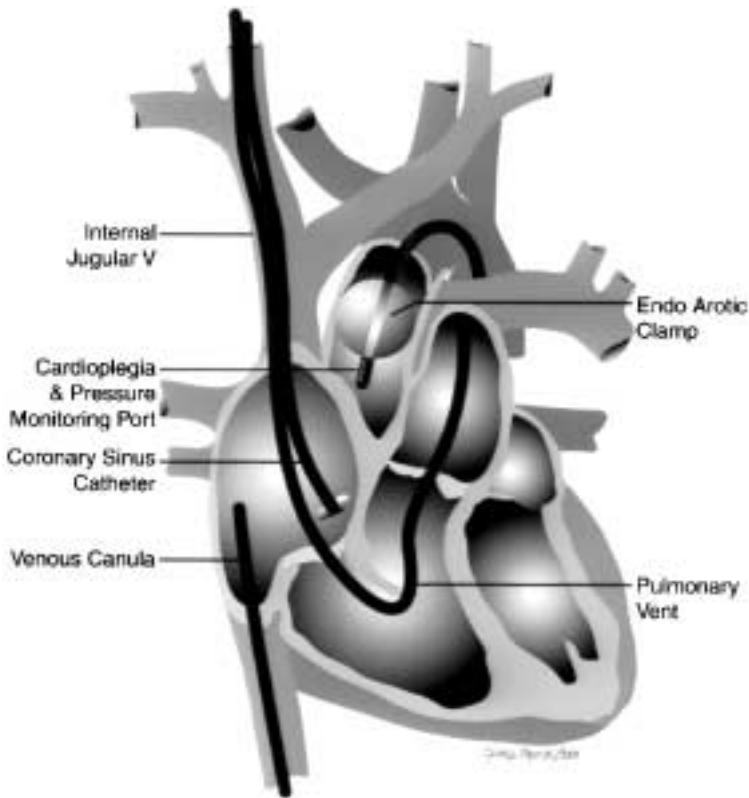


Figure 2. Port-access system for initiating percutaneous cardiopulmonary bypass. In patients with severe peripheral vascular disease, the endo-aortic balloon may be replaced by a directly placed endoclamp. The author wishes to acknowledge the help of George Moogk for fig.2.

anaesthesiologist familiar with the surgical procedure planned. Thoracic spine and the skin overlying the T1–T5 area should be examined for suitability of neuraxial techniques for post-operative analgesia. Airway and chest anatomy should be suitable for insertion of a double lumen tube to facilitate one-lung ventilation when required. Thus, patients with documented difficult intubation, scoliosis and emphysematous chest may be identified in the pre-operative clinic as unsuitable for this type of surgery. Although pulmonary function values have been reported not to correlate or predict post-operative ventilatory failure, patients with severe chronic obstructive pulmonary disease or asthma do not tolerate prolonged one-lung ventilation of the extent required for this type of surgery. Permissive hypercarbia has been used intra-operatively in patients undergoing lung surgery,²⁶ but may necessitate the use of inotropes and may cause ventricular irritability, both of which will be undesirable in a patient undergoing revascularization procedures on a beating heart. Pulmonary function tests (PFT), arterial blood gases and chest X-ray may provide the necessary information to identify patients who may not tolerate prolonged one-lung ventilation. Patients with resting hypercarbia (> 50 mmHg) and hypoxia ($\text{PaO}_2 < 60$ mmHg on room air), active bronchospasm, major emphysematous bullae, will not tolerate the additional hypercarbia and hypoxia resulting from prolonged one-lung ventilation and carbon dioxide insufflation. For those

with milder respiratory compromise, pulmonary function should be optimized with the aid of bronchodilators and physiotherapy. Patients should be encouraged to stop smoking at least 2 weeks before surgery. Similarly, morbid obesity, apart from making the surgical approach difficult, can pose problems with ventilation perfusion mismatch.

The potential for hypoxia, incomplete revascularization and haemodynamic changes poses significant risk to patients with diffuse multivessel coronary artery disease, congestive heart failure, poor left ventricular function, and major valvular lesions. Patients with unstable angina or recent myocardial infarction have the potential for myocardial irritability and dysfunction associated with recent ischaemia/injury. Access to the heart for managing life-threatening arrhythmia is limited. Intuitively, such patients are better managed with alternative techniques, such as OPCAB, as defibrillation and CPB can be initiated quickly if required, but currently there are no data on this.

Patients who have coronary angioplasty within 2 weeks prior to non-cardiac surgery have been documented to have an increased risk of peri-operative myocardial events and death.^{27,28} Although there are no studies looking at this in the minimally invasive cardiac surgical group, if a hybrid procedure is planned, it would be prudent not to do the angioplasty within 2 weeks prior to surgery. In our institution angioplasty for non-operated coronaries in the multi-vessel group is done during the post-operative angiography. Significant peripheral vascular disease may preclude port-access surgery, as initiating CPB via femoral vessels may pose unacceptable risk of limb ischaemia in this group.

Pre-operative medications, particularly beta blockade, should be continued through the peri-operative period. Elderly patients who are significantly beta blocked may not tolerate the additional sympathetic blockade from thoracic epidural analgesia (TEA) and may require transvenous pacing. Antiplatelet agents, such as clopidogrel, if combined with aspirin, should be stopped at least 5 days prior to surgery – particularly if neuraxial block is planned for analgesia as there are inadequate data on their propensity to cause neuraxial haematoma. Oral anticoagulants such as coumadin should be discontinued in adequate time. Patients requiring ongoing anticoagulation with heparin will not be suitable for neuraxial blocks. Coagulation parameters should be checked pre-operatively and they should be normal prior to initiation of neuraxial blocks. Discontinuing antiplatelet agents for the sake of neuraxial block should be done keeping in mind the risk of infarction in certain patients with recent infarct or unstable angina due to plaque rupture. Electrolytes such as potassium and magnesium should be within normal range prior to anaesthesia in order to avoid major dysrhythmias.

SPECIAL CONSIDERATIONS FOR ANAESTHESIA

Cardiovascular system

As the surgical procedure is done on a beating heart in a normothermic patient, myocardial protection becomes a challenge. Ischaemic pre-conditioning, which increases the resistance to ischaemic damage, has been used for myocardial protection if the target coronary vessel is not totally occluded.²⁹ Intracoronary shunts, while providing a bloodless field for anastomosis, may also provide some myocardial preservation. Perfusion of distal coronary artery via a separate pump (perfusion-assisted direct CAB),³⁰ which has been used for myocardial preservation in OPCAB, is technically impossible with MIDCAB. On the supply side, dilation of the coronary vessel and maintenance of adequate coronary filling pressure are important, while reduced myocardial contractility and irritability optimize myocardial oxygen demand.

Some of these patients may have 'non-critical' lesions in other vessels with areas of myocardium at risk. The induction of one-lung ventilation poses difficulties with ischaemia monitoring at a critical time (vide infra). Surgery on the right coronary via right minithoracotomy may necessitate significant dislocation of the heart and produce arrhythmia and major haemodynamic changes. Thus, provision for external transthoracic defibrillation must be made. Vagal stimulation,³¹ intravenous adenosine³² and intracoronary cholinergic and beta-blocking drugs with a pacemaker³³ have all been used to produce transient cardiac standstill to facilitate anastomosis. With the improvement in stabilizer designs and the availability of robotic and video assistance, there is less need for such aggressive heart rate manipulation.

All intravenous lines should be double-clamped shut if CPB is aided with suction on the venous line to avoid air emboli. De-airing at the end of CPB should be thoroughly verified with transoesophageal echocardiography (TEE). Carotid compression with steep Trendelenburg position may be used during declamping to reduce cerebral air emboli.

Respiratory system

Single-lung ventilation and induction of pneumothorax with carbon dioxide insufflation is required during the entire surgical procedure for MIDCAB and Totally Endoscopic Coronary Artery Bypass (TECAB). Single-lung ventilation can reduce cardiac output, increase pulmonary artery pressure and pulmonary vascular resistance³⁴ and produce hypoxia and hypercarbia. These effects may be exaggerated in patients who are obese and/or are chronic smokers. Prolonged one-lung ventilation (OLV) can cause significant atelectasis, oedema and ventilation perfusion mismatch well into the post-operative period. The mean duration of OLV in our series of 48 patients during MIDCAB was 196.8 ± 67 minutes which was doubled in patients undergoing robotically assisted surgery.³⁵ If the patient is bronchospastic or has significant pathology in the contralateral lung, delivering adequate ventilation to maintain eucardia may be difficult. Pneumothorax intermittently can become a tension pneumothorax, resulting in significant reduction in venous return and severe hypotension. This requires frequent adjustments in insufflation pressure as well as venting. Continuous monitoring of insufflation pressure, airway pressures, expired tidal volume and central venous pressures are required. Anaesthesiologist and surgeon should communicate with each other frequently in order to avoid unnecessary administration of vasoactive drugs. The hypoxia and hypercarbia per se elevate pulmonary artery pressure, pulmonary vascular resistance and reduce cardiac output,³⁴ which may confound ischaemia monitoring. Pneumothorax induces changes in the electrocardiographic axis and the amplitude of the waveform, further confounding ischaemia and arrhythmia monitoring.

While the majority of patients can be extubated at the end of surgery, 20% of patients may require short-term post-operative ventilation because of hypothermia, sedation or for not fulfilling the standard extubation criteria.³⁵ Use of double-lumen tubes for single-lung anaesthesia will necessitate change of this tube to a single-lumen endotracheal tube at the end of anaesthesia, requiring deeper levels of anaesthesia and paralysis. Use of the Univent[®] endotracheal tube with bronchial blocker makes short-term post-operative ventilation easier without the need for tube change at the end of anaesthesia. Insertion of the Univent tube is also less traumatic.

One-lung ventilation causes significant oliguria,³⁶ but once double-lung ventilation is resumed the urine output improves. Use of diuretics during this period causes unnecessary dehydration, resulting in postural hypotension in the post-operative

period apart from a potential for arrhythmia. Fluid overload, particularly with crystalloids, could potentially cause oedema in the collapsed lung, resulting in post-operative hypoxaemia. This may be worsened if conversion to CPB is required owing to the additional fluids from crystalloid pump prime.

Patient positioning and monitoring

Patients are positioned in the lateral tilt position with the arm suspended from a support at the level of the head. This can result in brachial plexus injury. Patients undergoing robotic assisted MIDCAB may have their arm free-draped to facilitate instrumentation, and additional care should be taken to avoid neurological injury. Patients are draped exposing thorax, abdomen, both groins and one lower limb for surgical access. This leaves only one arm for arterial pressure monitoring and intravenous access. Surgical incision anteriorly in the 4th or 5th intercostal space and ports in the midaxillary line preclude proper positioning of the lateral chest lead such as V5 and V6, confounding ischaemia monitoring in the left anterior descending (LAD) artery territory. Multiple monitoring modalities for ischaemia monitoring are essential. Transoesophageal echocardiography (TEE) is an excellent tool for evaluating ischaemia with regional wall motion abnormalities and left ventricular diastolic function assessment. Mehta et al³⁷ have noted the pitfalls with TEE in MIDCAB, but there are no data on the interaction between tension pneumothorax, hypercarbia and diastolic function evaluation as a measure of ischaemia. TEE is also used to verify continually the position of endoclamp and de-airing at the end of surgery, and to evaluate the function of the repaired or replaced valve. Transcranial Doppler (TCD) is a valuable tool for identifying endovascular aortic balloon clamp malposition.³⁸ Continuous mixed venous saturation monitoring with an oxymetric pulmonary artery catheter is an excellent tool for monitoring acute changes in oxygen supply demand match which integrates the effect of intermittent acute tension pneumothorax, hypoxaemia and alteration in cardiac output induced by one-lung ventilation and/or ischaemia. There are no studies documenting its efficacy compared to TEE in this subset of patients. Continuous cardiac output and right ventricular preload measuring devices are just undergoing clinical evaluation.

Neuromuscular block monitoring is important so that the patient does not cough during the process of delicate anastomosis through the port or the minithoracotomy. Infusion of a muscle relaxant allows some freedom to monitor and execute other events and therapies.

Accidental hypothermia during MIDCAB often occurs unless steps are taken to prevent it. Hypothermia can delay extubation or cause shivering in the post-operative period, increasing oxygen requirements significantly. Patients should not be extubated unless the patients' core temperature is above 36 degree Celsius in order to avoid post-operative shivering. We usually position the hose of the ambient air warmer under the drapes as there is no room for placing the conventional blanket. All intravenous fluids are warmed. Use of a circle system helps to humidify and warm the inspired air. The carbon dioxide absorber can become exhausted faster with increased CO₂ output from the insufflations and therefore needs frequent checking. External defibrillator pads are positioned prior to anaesthesia. The position of the pads will have to be modified depending on the surgical approach. In certain cases where pacing may be required, transvenous pacing wires are inserted along with the pulmonary artery catheters and checked for pacing at induction. All points of pressure should be appropriately padded and protected.

Post-operative pain

Surgical incision and ports might be spread over the T2–T9 dermatomal segments (Figure 1). Conversion to either partial or full sternotomy may be necessary to complete surgery if the coronary anatomy is unsuitable for approach via a minithoracotomy or there are complications such as cardiac perforation by stabilizer and port-access catheters. Thus the analgesic technique should be titrable. Clinical experience suggests that lateral thoracotomy is more painful than median sternotomy.³⁹ Clinically, these patients experience significant pain in the first 48 hours. This can result in sympathetic stimulation, increased stress response and release of catecholamines and other stress hormones. Pain also causes hypoventilation, resulting in hypoxia during a period of increased myocardial oxygen demand.

Anticoagulation

Unlike patients undergoing conventional CABG, MIDCAB patients receive smaller doses of heparin (10–20 000 units) with activated clotting time (ACT) aimed at 400 seconds. ACT should be checked frequently to ensure adequate anticoagulation. IMA is clamped and divided just before anastomosis, and the surgeon should make sure that IMA is not clotted prior to performing the anastomosis. Coagulation function is better maintained at the end of surgery as CPB is avoided in MIDCAB.

MANAGEMENT OF ANAESTHESIA

The management of anaesthesia should take into consideration the multiple factors affecting various systems and facilitate safe, smooth and timely recovery in order to accomplish the economic benefits of minimally invasive cardiac surgery. Patients can be pre-medicated with a small dose of benzodiazepine. Monitoring should include five-lead EKG (displaying lead II and 'V5') with computerized ST segment trending, invasive arterial pressure, oxymetric pulmonary artery catheter with thermodilution cardiac output, and/or transoesophageal echocardiography, continuous mixed venous saturation, TCD when indicated, temperature, airway pressure, expired tidal volume, hourly urine output, pulse oxymetric saturation, intermittent arterial blood gases and a neuromuscular blockade monitor. Thus, the surgery may be minimally invasive but anaesthetically these patients require extensive monitoring. External defibrillator pads are applied prior to induction of anaesthesia. Either a left-sided double-lumen tube or a single-lumen tube with bronchial blocker such as the Univent[®] tube is used to facilitate one-lung ventilation. Continuous positive airway pressure (CPAP) is applied to the collapsed lung to improve oxygenation and reduce shunt fraction. Occasionally, intermittent ventilation of the collapsed lung will be required to maintain oxygenation and to correct hypercarbia. Nitroglycerine is used to control ST segment changes or elevations of pulmonary artery diastolic pressure. Occasionally, correction of hypercarbia and hypoxia will be necessary to correct the changes in pulmonary artery pressures. Intravenous lidocaine and magnesium may be used prior to pre-conditioning and anastomosis to reduce ventricular irritability. With improvement in equipment design, techniques originally described to produce temporary cardiac standstill^{31–33} or to reduce heart rate for anastomosis, such as infusion of esmolol,⁴⁰ are less frequently used. Intra-operative cell salvage is kept as a standby. Careful attention should be paid to fluid balance. While adequate intravascular volume status is important

for haemodynamic stability and organ perfusion, conversion to cardiopulmonary bypass might result in crystalloid overload with the pump prime solution. Use of colloids may provide the advantage of reduction in pulmonary interstitial water. Initiation of controlled ventilation and one-lung anaesthesia and the pneumothorax can lead to a number of haemodynamic and hormonal changes such as reduced cardiac output, elevated pulmonary artery pressure, increased antidiuretic hormone (ADH), increased vasopressin³⁶ and reduced atrial natriuretic peptide levels. These changes result in reduced glomerular filtration rate and low urine output.

A number of anaesthetic techniques have been used for MIDCAB. Balanced anaesthesia using narcotics, muscle relaxant and inhalation agents or propofol infusion, is commonly used. Propofol has been documented to reduce pulmonary vascular resistance if the sympathetic tone is low. In the presence of lowered sympathetic tone resulting from TEA,^{41,42} propofol may contribute to improved oxygenation. Combination of TEA with general anaesthesia has been reported to be ideal for oxygenation during one-lung ventilation.⁴³ A variety of regional anaesthetic blocks or intrathecal morphine have been used to reduce narcotic requirements. Post-operative analgesia has been provided by patient-controlled analgesia (PCA) with narcotics, intercostal block,³⁵ wound infiltration,⁴⁴ interpleural block,⁴⁵ intrathecal morphine sulphate,⁴⁶ thoracic epidural analgesia^{45,47} or continuous percutaneous paravertebral block.^{48,49} The traditional narcotic-based anaesthesia using 25–50 µg/kg of fentanyl or its equivalent, used for conventional CABG, is obviously inappropriate as it necessitates post-operative ventilation for an average of 19 ± 1.5 hours. This could be shortened to 4.1 ± 1.3 hours with the use of lower doses of narcotics and propofol.⁵⁰ Use of nitroglycerine and/or nitroprusside, beta blockers and continuing sedation are required for achieving smooth transition to extubation, requiring admission of these patients to the intensive care unit. In our experience with MIDCAB, failure to meet standardized extubation criteria, such as EKG changes, sedation, hypothermia and poor pain control, are the principal causes of delay in extubation. Post-operative EKG changes and chest pain prolonged stay in the ICU beyond 24 hours. Following MIDCAB, 20–30% of patients may have EKG changes secondary to pericardial reaction in the absence of myocardial ischaemia. Incisional pain in the presence of EKG changes necessitates urgent angiographic or transthoracic Doppler evaluation of graft patency to rule out graft thrombosis. Thus, aggressive management of post-operative pain in this subset of patients is vital to facilitate fast-tracking.

Regional analgesic techniques

One of the disadvantages of narcotic-based anaesthesia is the sleep disturbance associated with the use of narcotics. Earlier studies in patients undergoing non-cardiac surgery have shown that the sleep disturbances associated with narcotics can cause sympathetic nervous system activity causing tachycardia, hypertension and ST segment changes that may play a role in peri-operative myocardial ischaemia and infarction.⁵¹ Considering the potential for incomplete revascularization with MIDCAB, narcotics may pose additional risks. Following thoracotomies, regional anaesthetic techniques have been documented to provide specific advantages such as reduction in intra- and post-operative narcotic requirement with improved analgesia during physical activity^{43,52} resulting in improved pulmonary function and early discharge. Local anaesthetic absorbed from these regional blocks may have beneficial anticoagulant effects, improving graft function.⁵³ A number of regional anaesthetic blocks have been used for minimally invasive cardiac surgery.

Blockade of multiple intercostal nerves

An extrapleural intercostal catheter under direct vision at the end of thoracotomy has been reported to improve pulmonary function and to provide excellent pain relief. Plasma concentrations of bupivacaine are as high as 4.5 µg/ml following intercostal blocks.⁵⁴ In animal experiments plasma levels of bupivacaine near 4 µg/ml can cause ventricular arrhythmia, particularly in the presence of ischaemia.^{55,56} Establishment of the block at the end of surgery by the surgeon necessitates deeper levels of anaesthesia and/or larger doses of narcotics intra-operatively for haemodynamic stability. The block cannot be predictably extended if the upper abdomen is used for portals or if conversion to sternotomy is required. There is incomplete sympathectomy peri-operatively, resulting in more heart rate variability, hypertension and the need for vasoactive drugs peri-operatively.³⁵ It is technically difficult if not impossible to strip adequately the parietal pleura posteriorly and to position the catheter optimally via the small anterior thoracotomy. Interpleural block⁴⁵ with injection of 20–25 ml of local anaesthetic into the interpleural space provides excellent pain relief but the chest tube has to be clamped for 30 minutes following interpleural injections. Local anaesthetic toxicity, infection, pneumothorax, pleural effusion and Horner's syndrome have all been described as complications resulting from the use of this technique.⁵⁷ Bilateral sympathetic block has been documented after interpleural block,⁵⁸ which might be beneficial.

Percutaneous continuous paravertebral block

Catheterization of the paravertebral space can be done percutaneously.^{48,49,59} At the paravertebral space the sympathetic chain is very close to the somatic nerves, and bilateral sympathetic block can occur with paravertebral blocks.⁶⁰ The block can be initiated just prior to surgery, thus reducing intra-operative anaesthetic requirements, and can provide cardiac sympathectomy. In a randomized study, Dhole et al⁴⁹ found it to be as effective as thoracic epidural analgesia for MIDCAB. The overall failure rate is about 10% (adults 10.7%; children 6.2%). Complications include hypotension (4.6%), vascular puncture (3.8%), pleural puncture (1.1%) and pneumothorax (0.5%).⁶¹ There can be accidental entry into the pleural cavity, in which case one may convert it to an interpleural block. There have been reports of pulmonary haemorrhage secondary to lung penetration in patients with a previous scar in the paravertebral area.⁶² Bilateral paravertebral blocks may be used if sternotomy is planned.

Thoracic epidural analgesia (TEA)

TEA has been the gold standard for post-thoracotomy analgesia.⁵² The risk of neuraxial haematoma with heparinization precluded its use in the setting of cardiac surgery until recently. The specific advantage of cardiac sympathectomy in patients with coronary artery disease^{63,64} has resulted in a resurgence of this technique in cardiac anaesthesia. The incidence of neuraxial haematoma may be 'estimated' to be 1 in 1000 and 1 in 3000, respectively, in the cardiac surgical patients as this complication has not been reported in this subset of patients.⁶⁵ Trauma to the epidural vein resulting in venous bleeding will be restricted as the epidural veins are collapsible when epidural pressure exceeds venous pressure. It is the rare arterial trauma in the epidural space that can continue to bleed in the presence of systemic anticoagulation, resulting in neurological injury. TEA offers specific advantages in MIDCAB patients compared to traditional

Table 2. Advantages of TEA for minimally invasive cardiac surgery compared to conventional balanced anaesthesia with or without intercostal block.

Effect	Advantage
Reduction in intra- and post-operative narcotic requirement	Less sedation, nausea, respiratory depression
Improved analgesia during activity, coughing	Early ambulation, improved pulmonary function
Extendable analgesia to bilateral and over several dermatomes	Adequate for conversion to sternotomy
Epicardial coronary dilation	Facilitate anastomosis via minimal access, increased anastomotic flow
Minimal heart rate variability, reduced myocardial irritability	Facilitate anastomosis
Improved endo- to epicardial flow	Ischaemia protection, faster recovery after an ischaemic episode
Reduced myocardial oxygen requirements	Myocardial protection particularly in hybrid procedures
Reduced intra-operative catecholamine and cortisol surge	Less stress response, less hyperglycaemia
Reduction in requirement of vasoactive drug infusions post-operatively	Reduction in ICU stay

balanced anaesthesia with or without intercostals block (Table 2). Coronary vasodilation resulting from TEA can facilitate anastomosis and increase anastomotic flow (43 ± 28 ml/minute versus 27 ± 7.7 ml/minute).⁶⁶ It allows smooth transition from surgical anaesthesia to post-operative analgesia and reduces intra-operative stress hormone release.

TEA may have certain disadvantages. Hypotension, particularly diastolic hypotension, due to sympathectomy can reduce coronary filling pressure – but the coronary dilation and reduced myocardial oxygen requirements offset such a problem. We accept a slightly lower blood pressure parameter provided that it is adequate for coronary and cerebral perfusion as correcting numbers with vasoactive drugs might be counterproductive with myocardial protection. Weakness of respiratory muscles has been documented by Sakura et al⁶⁷ to reduce resting minute and tidal ventilation by 10–15% without affecting ventilatory response to hypoxia and hypercarbia. The use of lower concentrations of local anaesthetics, such as 0.2% ropivacaine, does not produce muscle weakness, but higher concentrations used intra-operatively could cause motor weakness. Fortunately this occurs intra-operatively when these patients are ventilated. Epidural infusion of lower concentration of ropivacaine also allows neurological monitoring for epidural haematoma. Patients undergoing post-operative coronary angiogram may receive some heparin, and the removal of an epidural catheter should be done after establishing normal coagulation parameters.

TEA obtunds hypoxic pulmonary vasoconstriction (HPV) in the collapsed lung. Garutti et al⁶⁸ reported lower PaO_2 (120 ± 67 mmHg versus 175 ± 69) and increased shunt fraction during one-lung ventilation in the presence of TEA during pulmonary surgery, whereas this has not been a problem when TEA is used for conventional CABG.⁶⁹ Other investigators have documented better oxygenation with TEA combined with general anaesthesia for lung surgery.⁴³ The use of continuous positive airway pressure (CPAP) to non-dependent lung corrects this problem and may reduce microatelectasis. Although there are a number of studies looking at TEA for CABG and MIDCAB, there are no data on the use of TEA in minimally invasive valve surgery.

Postural hypotension in the post-operative period can interfere with ambulation. It is important to keep the patient adequately hydrated by replacing chest tube losses promptly. A sensory block between T1 and T8 should be established prior to induction of general anaesthesia in order to derive benefits from cardiac sympathectomy. The intra-operative narcotic dose should be limited if early extubation is planned. Recently, Karagos et al⁷⁰ have performed coronary artery bypass surgery using an H graft and lower ministernotomy in awake patients under TEA. NSAIDs and acetaminofen are important adjuncts to multimodal analgesia in these patients.

Intrathecal opioid

Intrathecal morphine sulphate (ITMS) under $10 \mu\text{g/kg}$ in CABG has been devoid of significant sedation but offers significant analgesia and reduces narcotic requirement in the peri-operative period.⁷¹ There are no reports of randomized studies with safety and efficacy of ITMS for MIDCAB, whereas it is used for regular CABG and off-pump CABG (OPCAB) in several centres. It has been documented to result in increased episodes of sedation and hypoxaemia and therefore does not obviate the undesirable side-effects of opioid.

LOGISTICAL PROBLEMS AND FAST-TRACKING

In many institutions patients for minimally invasive cardiac surgery are admitted to the hospital on the day of surgery and are usually discharged between 48 and 72 hours post-operatively. This poses a problem with the initiation of neuraxial block as well as re-scheduling another patient if the epidural insertion results in vascular trauma. Although it might be safe to insert the catheter until 1 hour prior to heparinization, postponing surgery if a blood vessel is accidentally entered, particularly if the blood looks arterial, poses problems with scheduling another patient. We insert the catheter the evening before surgery as an outpatient procedure and send patients home or to a local hotel with the catheter securely taped. This gives us ample time for catheter insertion, facilitates teaching participation and enables re-scheduling another patient should this become necessary due to bleeding. Monitoring following epidural catheter removal has to be done by telephone call/home care. There are recent studies with TEA with a small sample size that report no neuraxial haematoma. Currently we do not know the exact risk of neuraxial haematoma with TEA in this subset of patients. This is one of the reasons why paravertebral block is gaining popularity, even if one has to do bilateral blocks.

Although patients following lung surgery with significant coronary artery disease have bypassed ICU for a number of years, particularly with the use of thoracic epidural analgesia, the combination of thoracotomy with cardiac surgery creates a natural concern in the nursing staff, both in post anaesthesia care unit (PACU) and in the regular wards. These patients need to be closely monitored for anastomotic patency, myocardial ischaemia and infarction, arrhythmia and post-operative bleeding; an optimal nurse/patient ratio is important for effective monitoring. A standardized

Table 3. Extubation criteria at the end of surgery.
Awake, cooperative patient
Adequate analgesia
Adequate ventilation:
Pulse oxymetric saturation > 95% on FiO ₂ of < 0.5
PaCO ₂ < 55 mmHg
Tidal volume at least 5 ml/kg
Vital capacity > 10 ml/kg
PaO ₂ > 80 mmHg on FiO ₂ 0.5
Temperature > 36.5°C
Acid–base status:
pH > 7.3
Potassium > 4.0 mmol/l
Chest tube drain < 50 ml/30 minutes
Absence of arrhythmia:
No SVT
No ventricular irritability
Haemodynamic stability:
Cardiac index > 2.0 l/minute/m ⁻²
No inotropic requirement
No pacemaker requirement
Nausea-free

Practice points

- minimally invasive cardiac surgery is evolving. The technology and equipment are improving everyday. Currently, the duration of surgery is significantly prolonged compared to traditional cardiac surgery
- incomplete re-vascularization is still a major concern with minimally invasive coronary bypass surgery
- monitoring for ischaemia should be multimodal
- access to the heart is limited, and external defibrillators and pacemakers must be readily available
- analgesic techniques should be extendable to cover sternotomy if conversion to sternotomy is required. TEA and paravertebral block offer specific advantages
- prolonged single-lung ventilation can pose problems with post-operative oxygenation
- maintaining normothermia and normal acid–base status is essential to facilitate early extubation
- initiation of CPB using femoral vessels as well as percutaneously introduced catheters pose additional risks such as air entrainment in the venous system as well as trauma to coronary sinus and right ventricle
- TEE is a valuable tool for facilitating percutaneous insertion and for the positioning of various canulae as well as for verifying adequate de-airing at the end of CPB following minimally invasive valve surgery

Research agenda

- prospective studies are required to evaluate the role of anaesthetic techniques on cost and outcomes following minimally invasive surgery
- pulmonary function after such prolonged single-lung ventilation should be evaluated. The role of analgesic techniques in modulating pulmonary dysfunction needs evaluation in this subset of patients
- quality-of-life outcomes following minimally invasive cardiac surgery using different anaesthetic and analgesic techniques should be evaluated
- prospective randomized trials are needed to evaluate the true benefits of such intervention. Currently there is a lot of selection bias with minimally invasive cardiac surgery. There is a need to evaluate the role of minimally invasive surgery in redo operations
- studies are required to see whether one can bypass the intensive care unit following minimally invasive surgery

protocol regarding suitability for fast-tracking, extubation criteria, criteria for transfer out of PACU and management of patients who fail to fit these criteria should be developed to manage these patients safely (Table 3). Outcome studies are required to establish the value of such surgery and anaesthetic care. Currently patients undergoing minimally invasive cardiac surgery seem to be pre-selected. Unless a similar group is prospectively compared with traditional surgical technique, the results published so far on minimally invasive cardiac surgery are tainted with selection bias.

CONCLUSION

In order to optimize the benefits of minimally invasive cardiac surgery, anaesthesia and analgesia have to be tailored appropriately. Patient selection is very important for fast-tracking these patients. Regional anaesthesia and analgesia with TEA or paravertebral block offers the ideal conditions for realizing the economic benefits of minimally invasive surgery. Intrathecal morphine and interpleural blocks also offer good analgesia but need prospective evaluation.

REFERENCES

1. Zenati M, Domit TM, Saul M et al. Resource utilization for minimally invasive direct and standard coronary artery bypass grafting. *Annals of Thoracic Surgery* 1997; **63**: S84.
2. Reichenspurner H, Boehm D, Detter C et al. Economic evaluation of different minimally invasive procedures for the treatment of coronary artery disease. *European Journal of Cardiothoracic Surgery* 1999; **16** (supplement 2): S76–S79.
3. Cohn LH. Minimally invasive aortic valve surgery: technical considerations and results with the parasternal approach. *Journal of Cardiac Surgery* 1998; **13**: 302–305.
4. Byrne JG, Mitchell ME, Adams DH et al. Minimally invasive direct access mitral valve surgery. *Seminars in Thoracic and Cardiovascular Surgery* 1999; **11**: 212–222.
5. Colvin SB, Galloway AC, Ribakove G et al. Port-access mitral valve surgery: summary of results. *Journal of Cardiac Surgery* 1998; **13**: 286–289.
6. Frazier BL, Derrick MJ, Purewal SS et al. Minimally invasive aortic valve replacement. *European Journal of Cardiothoracic Surgery* 1998; **14** (supplement 1): S122–S125.
7. Subramanian VA, McCabe JC & Geller CM. Minimally invasive direct coronary artery bypass grafting: two year clinical experience. *Annals of Thoracic Surgery* 1997; **64**: 1648–1653.
8. Wittwer T, Cremer J, Boonstra P et al. Myocardial 'hybrid' revascularisation with minimally invasive direct coronary artery bypass grafting combined with coronary angioplasty: preliminary results of a multicentre study. *Heart* 2000; **83**: 58–63.
- * 9. Casey LC. Role of cytokines in the pathogenesis of cardiopulmonary induced multisystem organ failure. *Annals of Thoracic Surgery* 1993; **56**: S92–S96.
10. Tonnesen E, Brix Christensen V & Toft P. The role of cytokines in cardiac surgery. *International Journal of Cardiology* 1996; **53** (supplement): 1–10.
11. McBride WT, Armstrong MA, Crockard AD et al. Cytokine balance and immunosuppressive changes in cardiac surgery: contrasting response between patients and isolated CPB circuits. *British Journal of Anaesthesiology* 1995; **75**: 724–733.
12. Ganapathy S, Murkin JM, Dobkowski W et al. Cytokine response during minimally invasive coronary artery bypass (MIDCAB) surgery: Role of anesthetic technique. Canadian Cardiovascular Society Annual Meeting, December 2000; abstract 554.
13. Murkin JM, Boyd WD, Ganapathy S et al. Beating heart surgery: why expect less central nervous system morbidity? *Annals of Thoracic Surgery* 1999; **68**: 1498–1501.
14. Nataf P, Lima L, Regan M et al. Thoracoscopic internal mammary artery harvesting: technical considerations. *Annals of Thoracic Surgery* 1997; **63**: S104–S106.
15. Subramanian VA & Patel NU. Transabdominal minimally invasive direct coronary artery bypass grafting (MIDCAB). *European Journal of Cardiothoracic Surgery* 2000; **17**: 485–487.
16. Cremer J, Struber M, Wittwer T et al. Off-bypass coronary bypass grafting via minithoracotomy using mechanical epicardial stabilization. *Annals of Thoracic Surgery* 1997; **63**: S79–S83.
- * 17. Grundeman PF, Borst C, van Herwaarden JA et al. Hemodynamic changes during displacement of the beating heart by the Utrecht Octopus method. *Annals of Thoracic Surgery* 1997; **63**: S88–S92.
18. Byrne JG, Karavas AN, Cohn LH & Adams DH. Minimal access aortic root, valve and complex ascending aortic surgery. *Current Cardiology Reports* 2000; **2**: 549–557.
19. Reichenspurner H, Weltz A, Guliemos V et al. Port-access cardiac surgery using endovascular cardiopulmonary bypass. *Journal of Cardiac Surgery* 1999; **14**: 275–280.
20. Applebaum RM, Colvin SB, Galloway AC et al. The role of transesophageal echocardiography during port-access minimally invasive cardiac surgery: a new challenge for the echocardiographer. *Echocardiography* 1999; **16**: 595–602.

- *21. Chaney MA. Anaesthetic management of port-access minimally invasive cardiac surgery. *Journal of the Indian Medical Association* 1999; **97**: 425–427.
22. Siegel LC, St Goar FG, Stevens JH et al. Monitoring considerations for port-access cardiac surgery. *Circulation* 1997; **96**: 562–568.
23. Grossi EA, Zakow PK, Ribakove G et al. Comparison of post-operative pain, stress response, and quality of life in port access vs. standard sternotomy coronary bypass patients. *European Journal of Cardiothoracic Surgery* 1999; **16** (supplement 2): S39–S42.
- *24. Chaney MA, Durazo-Arvizu RA, Fluder EM et al. Port-access minimally invasive cardiac surgery increases surgical complexity, increases operating room time, and facilitates early postoperative hospital discharge. *Anesthesiology* 2000; **92**: 1637–1645.
25. Abramson DC & Giannotti AG. Perforation of the right ventricle with a coronary sinus catheter during preparation for minimally invasive cardiac surgery. *Anesthesiology* 1998; **89**: 519–521.
26. Morisaki H, Serita R, Innami Y et al. Permissive hypercapnia during thoracic anesthesia. *Acta Anaesthesiologica Scandinavica* 1999; **43**: 845–849.
27. Kaluza GL, Joseph J, Lee JR et al. Catastrophic outcomes of noncardiac surgery soon after coronary stenting. *Journal of the American College of Cardiology* 2000; **35**: 1288–1294.
28. Posner KL, van Norman GA & Chan V. Adverse cardiac outcomes after noncardiac surgery in patients with prior percutaneous transluminal coronary angioplasty. *Anesthesia and Analgesia* 1999; **89**: 553–560.
29. Hawaleshka A & Jacobsohn E. Ischemic preconditioning: mechanisms and potential clinical applications. *Canadian Journal of Anesthesiology* 1998; **45**: 670–682.
30. Steele M & Palmer-Steele C. Perfusion technique for perfusion-assisted direct coronary artery bypass (PADCAB). *The Journal of Extra-Corporeal Technology* 2000; **32**: 158–161.
31. Matheny R & Shaar CJ. Vagus nerve stimulation as a method to temporarily slow or arrest the heart. *Annals of Thoracic Surgery* 1997; **63**: S28–S29.
32. Robinson MC, Theilmair KA & Hill BB. Transient ventricular asystole using adenosine during minimally invasive and open sternotomy coronary artery bypass grafting. *Annals of Thoracic Surgery* 1997; **63**: S30–S34.
33. Sakwa M, McCue MG, Shannon E et al. Pharmacologic electrical arrest with intermittent pacing facilitates off pump bypass surgery. *Second Annual Meeting and Scientific Sessions of the International Society for Minimally Invasive Cardiac Surgery (ISMICS)* 1999; **C18**: 33 (abstract).
34. Boldt J, Papsdorf M, Uphus D et al. Changes in regulators of the circulation in patients undergoing lung surgery. *British Journal of Anaesthesiology* 1997; **79**: 733–739.
35. Ganapathy S, Boyd WD, Morgan J et al. Thoracic Epidural Analgesia offers improved postoperative analgesia following MIDCAB. *Second Annual Meeting of the ISMICS* 1999; **C124**: 31 (abstract).
36. Golledge J & Goldstraw P. Renal impairment after thoracotomy: incidence, risk factors and significance. *Annals of Thoracic Surgery* 1994; **58**: 524–528.
37. Mehta Y, Juneja R & Dhole S. Transesophageal echocardiography in MIDCAB: pitfalls. *Journal of Cardiothoracic and Vascular Anesthesiology* 1999; **13**: 115–116.
38. Grocott HP, Smith MS, Glower DD & Clements FM. Endovascular aortic balloon clamp malposition during minimally invasive cardiac surgery: detection by transcranial doppler monitoring. *Anesthesiology* 1998; **88**: 1396–1399.
39. Diegeler A, Walther T, Metz S et al. Comparison of MIDCAP versus conventional CABG surgery regarding pain and quality of life. *Heart Surgery Forum* 1999; **2**: 290–295.
40. Alekscic I, Buhre W, Baryalei MM et al. Haemodynamic changes during minimally invasive coronary artery bypass surgery using high-dose esmolol. *Cardiovascular Surgery* 2000; **8**: 204–207.
41. Kondo U, Kim S, Nakayama M & Murray PA. Pulmonary vascular effects of propofol at baseline, during elevated vasomotor tone, and in response to sympathetic α - and β -adrenoreceptor activation. *Anesthesiology* 2001; **94**: 815–823.
42. Abe K, Shimizu T, Takashina M et al. The effects of propofol, isoflurane, and sevoflurane on oxygenation and shunt fraction during one-lung ventilation. *Anesthesia and Analgesia* 1998; **87**: 1164–1169.
43. von Dossow V, Welte M, Zaune U et al. Thoracic epidural anesthesia combined with general anesthesia: the preferred anesthetic technique for thoracic surgery. *Anesthesia and Analgesia* 2001; **92**: 848–854.
44. Borges MF & Coulson AS. Minimally invasive coronary bypass surgery: postoperative pain management using intermittent bupivacaine infiltration. *British Journal of Anaesthesiology* 1998; **80**: 519–520.
45. Mehta Y, Swaminathan M, Mishra Y & Trehan N. A comparative evaluation of interpleural and thoracic epidural analgesia for postoperative pain relief after minimally invasive direct coronary artery bypass surgery. *Journal of Cardiothoracic and Vascular Anesthesiology* 1998; **12**: 162–165.
46. Heres EK, Marquez J, Malkowski MJ et al. Minimally invasive direct coronary artery bypass: anesthetic, monitoring, and pain control considerations. *Journal of Cardiothoracic and Vascular Anesthesiology* 1998; **12**: 385–389.

47. Liem TH, Williams JP, Hensens AG et al. Minimally invasive direct coronary artery bypass procedure using a high thoracic epidural plus general anesthetic technique. *Journal of Cardiothoracic and Vascular Anesthesiology* 1998; **12**: 688–672.
48. Ganapathy S, Murkin JM, Boyd WD et al. Continuous percutaneous paravertebral block for minimally invasive cardiac surgery. *Journal of Cardiothoracic and Vascular Anesthesiology* 1999; **13**: 594–596.
49. Dhole S, Mehta Y, Saxena H et al. Comparison of continuous thoracic epidural and paravertebral blocks for postoperative analgesia after minimally invasive direct coronary artery bypass surgery. *Journal of Cardiothoracic and Vascular Anesthesiology* 2001; **15**: 288–292.
50. Cheng DCH. Fast-track cardiac surgery: economic implications in postoperative care. *Journal of Cardiothoracic and Vascular Anesthesiology* 1998; **12**: 72–79.
51. Rosenberg J, Rasmussen V, von Jessen F et al. Late postoperative episodic and constant hypoxemia and associated ECG abnormalities. *British Journal of Anaesthesia* 1990; **65**: 684–691.
52. Perttunen K, Nilsson E, Heinonen J et al. Extradural, paravertebral and intercostal nerve blocks for post thoracotomy pain. *British Journal of Anaesthesia* 1995; **75**: 541–547.
53. Kohrs R, Hoenemann CW, Feirer N et al. Bupivacaine inhibits whole blood coagulation in vitro. *Regional Anesthesia and Pain Medicine* 1999; **24**: 326–330.
54. Berrisford RG, Sabanathan S, Mearns AJ et al. Plasma concentrations of bupivacaine and its enantiomers during continuous extrapleural intercostal nerve block. *British Journal of Anaesthesiology* 1993; **70**: 201–204.
55. Freysz M, Timour Q, Bertrix L et al. Enhancement of ischemia of the risk of cardiac disorders, especially fibrillation, in regional anesthesia with bupivacaine. *Acta Anaesthesiologica Scandinavica* 1993; **37**: 350–356.
56. Picard S, Rouet R, Flais F et al. Proarrhythmic and antiarrhythmic effects of bupivacaine in an in vitro model of myocardial ischemia and reperfusion. *Anesthesiology* 1998; **88**: 1318–1329.
57. Stromskag KE, Minor B & Steen PA. Side effects and complications related to interpleural analgesia: an update. *Acta Anaesthesiologica Scandinavica* 1990; **34**: 473–477.
58. Ramajoli F & De Amici D. Is there a bilateral block of the thoracic sympathetic chain after unilateral interpleural analgesia? *Anesthesia and Analgesia* 1998; **87**: 360–367.
- *59. Richardson J & Lonnqvist PA. Thoracic paravertebral block. *British Journal of Anaesthesiology* 1998; **81**: 230–238.
60. Tenicela R & Pollan SB. Paravertebral-peridural block technique: a unilateral thoracic block. *Clinical Journal of Pain* 1990; **6**: 227–234.
61. Lonnqvist PA, MacKenzie J, Soni AK & Conacher ID. Paravertebral blockade. Failure rate and complications. *Anaesthesia* 1995; **50**: 813–815.
62. Thomas PW, Sanders DJ & Berrisford RG. Pulmonary hemorrhage after percutaneous paravertebral block. *British Journal of Anaesthesiology* 1999; **83**: 668–669.
63. Blomberg S, Emanuelsson H, Kvist H et al. Effects of thoracic epidural anesthesia on coronary arteries and arterioles in patients with coronary artery disease. *Anesthesiology* 1990; **78**: 840–847.
64. Kirno K, Friberg P, Grzegorzczak A et al. Thoracic epidural anesthesia during coronary artery bypass surgery: effects on cardiac sympathetic activity, myocardial blood flow and metabolism, and central hemodynamics. *Anesthesia and Analgesia* 1994; **79**: 1075–1081.
65. Ho AM, Chung DC & Joynt GM. Neuraxial blockade and hematoma in cardiac surgery: estimating the risk of a rare adverse event that has not (yet) occurred. *Chest* 2000; **117**: 551–555.
66. Ganapathy S, Dobkowski WB, Murkin JM et al. Thoracic epidural analgesia (TEA) increases graft flow and reduces cortisol release in minimally invasive coronary bypass surgery (MIDCABG). *Anesthesiology* 1999; **91**: A130.
67. Sakura S, Saito Y & Kosaka Y. The effects of epidural anesthesia on ventilatory response to hypercapnia and hypoxia in elderly patients. *Anesthesia and Analgesia* 1996; **82**: 306–311.
68. Garutti I, Quintana B, Olmedilla L et al. Arterial oxygenation during one lung ventilation: Combined versus general anesthesia. *Anesthesia and Analgesia* 1999; **88**: 494–499.
69. Tenling A, Joachimsson PO, Tyden H et al. Thoracic epidural anesthesia as an adjunct to general anesthesia for cardiac surgery: effects on ventilation-perfusion relationships. *Journal of Cardiothoracic and Vascular Anesthesiology* 1999; **13**: 258–264.
70. Karagos HY, Sonmez B, Bakkaloglu B et al. Coronary artery bypass grafting in the conscious patient without endotracheal general anesthesia. *Annals of Thoracic Surgery* 2000; **70**: 91–96.
- *71. Chaney MA. Regional anesthesia and cardiac surgery. *Anesthesia and Analgesia* 1997; **84**: 1211–1221.